Real-Time Video Generation with MiniMax H3

8× RTX PRO 5000 · vLLM-Omni + ComfyUI

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Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni + ComfyUI Applications

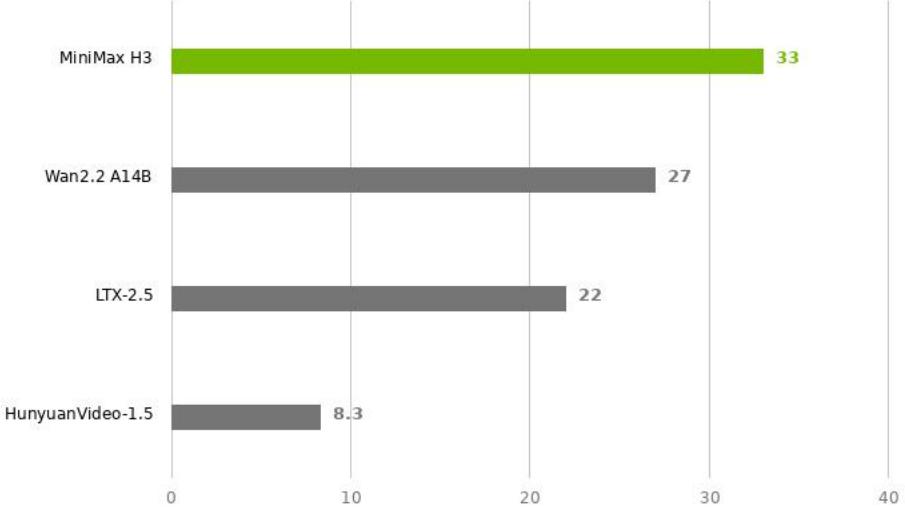


01 · Real-Time H3 Generation



Real-Time 768p Raises the Systems Bar

For 362 frames at 24 FPS, real time requires end-to-end latency < 15.1 s (RTF < 1.0)



GENERATIVE BACKBONE · FULL PARAMETERS (B)

RTF = latency / media duration. The real-time deadline here is 15.1 s. H3 is the largest dense backbone in this peer set; Wan2.2 = 27B total / 14B active.



OPEN H3-BASE · NATIVE 768P · 33B DENSE

32B encoder → 33B dense Omni Transformer → Visual/Audio VAE. 50 blocks × 50 denoising steps; full attention. ≈130 GB raw BF16; ~13B AdaLN cacheable.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 362-frame workload

Frontier Models · The Cards

Every discrete SM120 card combines fifth-generation Tensor Cores with FP4 and GDDR7; memory capacity and PCIe-to-fabric bandwidth define the deployment envelope

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

×8

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Baseline Breakdown

Exact 1280×720 request · 605.348 s E2E · DiT 584.707 s (96.6%) · 49 updates
Nsight reference · 1344×768 / 362F · first DiT step · GPU 0 kernel-sum

GPU kernel family	GPU0 ms / step	Kernel-time share	What it tells us
Dense attention	9,190.1	69.3%	50 main cuDNN SDPA calls Primary target → VSA / SageAttention
GEMM	2,693.1	20.3%	MLP + projections Target → distilled calls + low precision
NCCL communication	1,147.5	8.6%	SP8 SendRecv residency* Target → overlap / RDMA
Layout / index / copy	152.9	1.2%	Packing, transpose and device copies
Other / metadata	84.3	0.6%	Norm, activation and selection metadata
Categorized total	13,267.9	100.0%	GPU0 kernel-sum; not additive request wall time

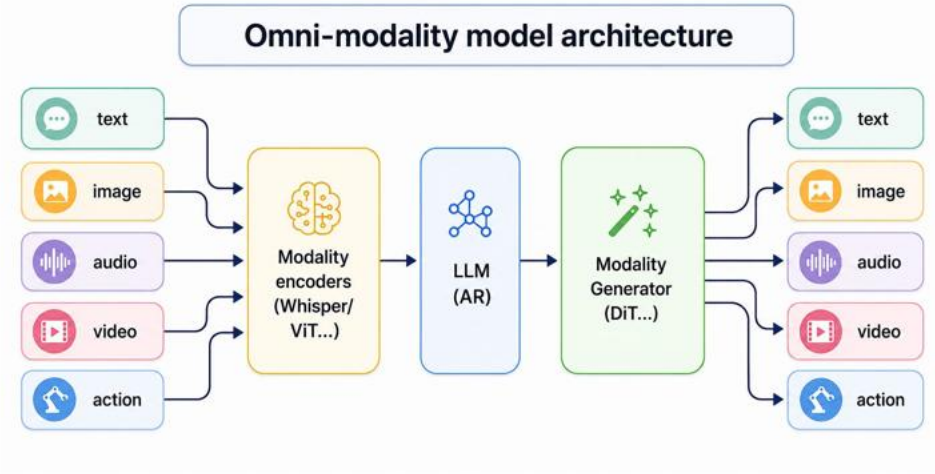
Author measurements · Nsight: 8× RTX PRO 5000, BF16, TP1/SP8, dense cuDNN attention, NCCL · kernel sums may overlap

02 · vLLM-Omni ComfyUI Apps



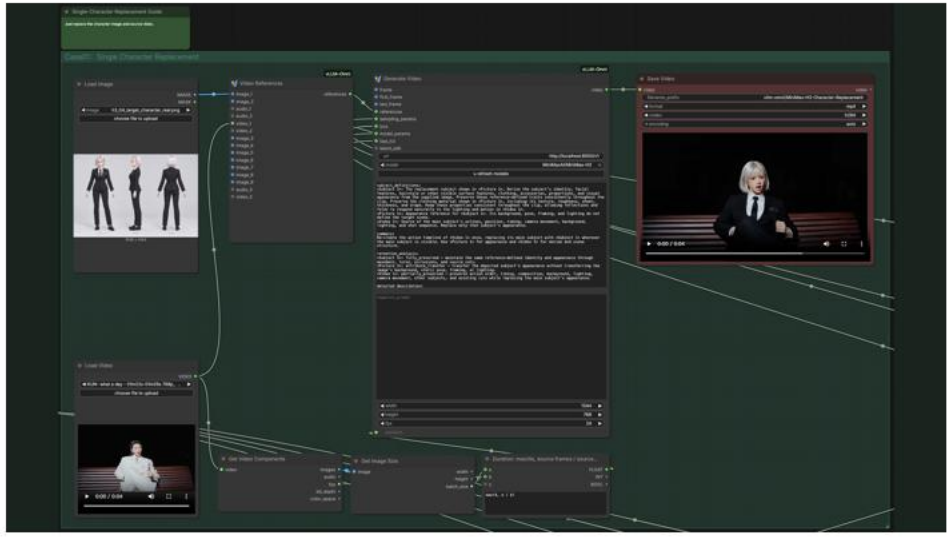
vLLM-Omni Connects ComfyUI to Multi-GPU H3 Serving

Open PR #9194: character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI remains the visual frontend; vLLM-Omni owns model execution, scheduling, and multi-GPU deployment



ComfyUI Workflow · PR #9194

Single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)